

THE FRACTURE MECHANICS OF DELAMINATION TESTS

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1 INTRODUCTION

The use of fracture mechanics to analyse the failure of fibre reinforced composites has been treated with some suspicion in the past. This arises from the correct perception that fracture mechanics is a rigorous discipline and the incorrect deduction that it is thus unsuited to dealing with the often complicated failure processes in composites. Fracture mechanics has, quite reasonably, been developed for simple fractures in homogeneous, isotropic materials, but the physics inherent in its methods are equally applicable to failures in inhomogeneous, anisotropic materials such as composites. Inhomogeneity is a problem whenever it occurs since it adds an extra characterisation requirement, but much can be done by analysing cracks within such systems. For instance, the behaviour of cracks at, or close to, interfaces, can give much insight into important variables such as interface adhesion and relative matrix-fibre properties. It is, of course, essential to analyse the problem on the correct scale so that the analysis can be locally homogeneous or at some defined boundary. Anisotropy is not a major difficulty since the analysis can be derived with equal rigour, albeit in a rather more complicated form.

This presence of anisotropy is an example of unexpected benefits which can arise to simplify the analysis. Highly anisotropic systems almost always delaminate, so the fracture plane is known and, for example, mixed mode loadings can be analysed without the uncertainties of unknown crack trajectories. In addition, composites are usually designed to be stiff, elastic materials, so they can often be treated as linearly elastic and the major difficulties of plastic deformation are avoided. It is early days in the subject, but fracture mechanics has found wide acceptance as a method of characterising the toughness of laminates in terms of their delamination behaviour. This is their weakest link and, hence, much effort has been expended on developing new, tough matrix materials to give high performance structures. It is the function of fracture mechanics to provide a rigorous basis for characterising, and thus comparing, these materials. The experience and methods of analysis used has produced some very precise tests, and these will form the basis of design methods. This paper will now describe how fracture mechanics has been used to define toughness in laminates and indicate how it has been applied.

2 LAMINATES UNDER BENDING

Let us consider just the case of the laminate shown in Fig. 1 which is of thickness $2h$ and has a delamination

crack through the thickness B a distance h_1 from the upper surface. If we suppose a general loading of M_1 on h_1 and M_2 on h_2 ($h_1 + h_2 = 2h$) then, by considering the energy change when the delamination grows (1)†, an expression for the total energy release rate can be derived as:

$$G = \frac{3}{4B^2hE_{11}} \left\{ \left(\frac{2h}{h_1} \right)^3 M_1^2 + \left(\frac{2h}{h_2} \right)^3 M_2^2 - (M_1 + M_2)^2 \right\} \quad (1)$$

where E_{11} is the axial modulus of the laminate as shown in Fig. 1. This can be partitioned into an opening, or mode I, component and one due to sliding or shear, mode II, by splitting the moments into

$$M_I = M_2 - \left(\frac{h_2}{h_1} \right)^3 M_1$$

and

$$M_{II} = M_2 + M_1$$

giving

$$G_I = \frac{6}{B^2E_{11}} \cdot \frac{h_1^3}{h_2^3(h_1^3 + h_2^3)} \left\{ M_2 - \left(\frac{h_2}{h_1} \right)^3 M_1 \right\}^2 \quad (2)$$

and

$$G_{II} = \frac{6}{B^2E_{11}} \cdot \frac{3h_1h_2}{(h_1 + h_2)^2(h_1^3 + h_2^3)} \cdot (M_2 + M_1)^2$$

When the moments are split in this way $G = G_I + G_{II}$ with no cross product terms.

In general one would expect G_{Ic} and G_{IIc} to be different since the deformation mode is different and there is no way, a priori, of knowing the energy requirement for mixed mode loading when both modes are present. A

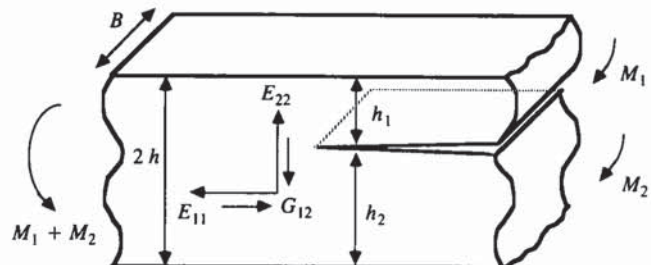


Fig. 1. Delamination under general bending

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† References are given in Appendix 2.

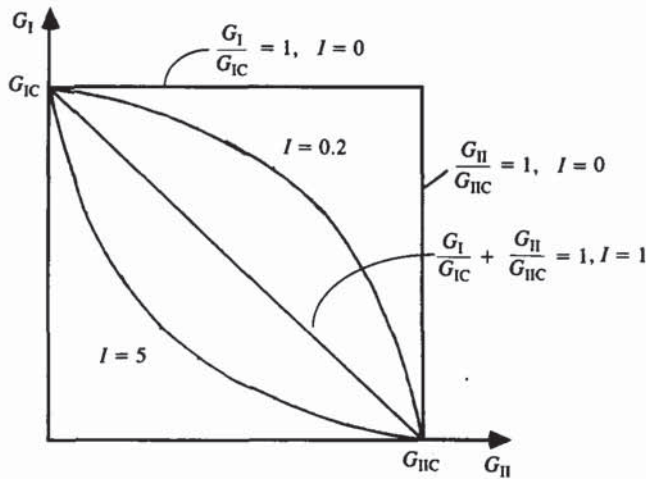


Fig. 2. Laminate failure locus

primary objective of characterising the laminate then is to determine the locus for failure as sketched in Fig. 2 of G_I versus G_{II} . On each axis we have the pure loading conditions of G_{IC} and G_{IIC} and various forms of possible mixed mode criteria; i.e., non-interactive

$$G_I/G_{IC} = 1 \text{ or } G_{II}/G_{IIC} = 1$$

or simple linear addition

$$\frac{G_I}{G_{IC}} + \frac{G_{II}}{G_{IIC}} = 1$$

These may be written as part of a general criterion

$$\left(\frac{G_I}{G_{IC}} - 1\right)\left(\frac{G_{II}}{G_{IIC}} - 1\right) - I \cdot \frac{G_I}{G_{IC}} \cdot \frac{G_{II}}{G_{IIC}} = 0 \quad (3)$$

Where I is an interaction parameter such that $I = 0$ gives the zero interaction result and $I = 1$ the simple addition. Lines are drawn in Fig. 2 for weak interaction, $I = 0.2$, and strong interaction, $I = 5$. It may also happen that the locus changes with amount of crack growth, the 'R curve' effect, and this may be represented as different lines for different a values.

In order to measure the loci described above it is necessary to have test geometries capable of producing various G_I/G_{II} conditions and the three shown in Fig. 3 have been found to be convenient. In Fig. 3(a) we have the well known double cantilever beam (DCB) specimen. The loading gives $M_2 = -M_1 = Pa$, so that from equations (2) we have

$$G_I = \frac{6P^2a^2}{B^2E_{11}} \cdot \left(\frac{1}{h_1^3} + \frac{1}{h_2^3}\right) \text{ and } G_{II} = 0 \quad (4)$$

i.e., a pure mode I test. In Fig. 3(b) we have the end loaded split test (ELS) for which

$$M_2 = Pa \left(\frac{h_2^3}{h_1^3 + h_2^3}\right) \text{ and } M_1 = Pa \left(\frac{h_1^3}{h_1^3 + h_2^3}\right)$$

Hence, from equation (2)

$$G_I = 0 \text{ and } G_{II} = \frac{6P^2a^2}{B^2E_{11}} \cdot \frac{3h_1h_2}{(h_1 + h_2)^2(h_1^3 + h_2^3)} \quad (5)$$

i.e., a pure mode II test. In Fig. 3(c) we have the

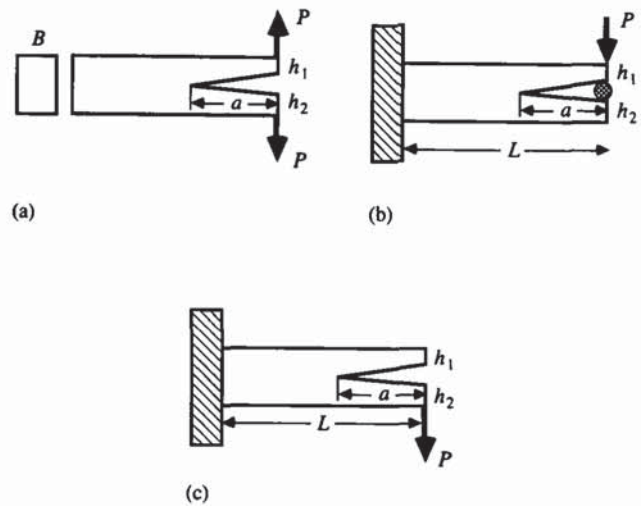


Fig. 3. Testing geometries

(a) The double cantilever beam test - DCB; pure Mode I

(b) The end loaded split test - ELS; pure Mode II

(c) The asymmetric end loaded split test - AELS; fixed ratio mixed mode

asymmetric end loaded split test (AELS) in which one arm is not loaded giving

$$M_1 = 0 \text{ and } M_2 = Pa$$

and hence

$$G_I = \frac{6P^2a^2}{B^2E_{11}} \cdot \frac{h_1^3}{h_2^3(h_1^3 + h_2^3)} \text{ and}$$

$$G_{II} = \frac{6P^2a^2}{B^2E_{11}} \cdot \frac{3h_1h_2}{(h_1 + h_2)^2(h_1^3 + h_2^3)} \quad (6)$$

giving a constant ratio test

$$\frac{G_I}{G_{II}} = \frac{1}{3} \left(\frac{h_1}{h_2}\right)^2 \left(1 + \frac{h_1}{h_2}\right)^2 \quad (7)$$

and hence radial lines on the G_I, G_{II} diagram.

3 CORRECTION FACTORS

The analysis given so far is based on simple beam theory and this is generally very accurate because many laminates can be tested as elastic, slender beams as required by beam theory. However, several factors arising from the particular nature of the specimens require special attention. Firstly, tough composites tested as slender beams often undergo large displacements during crack growth which lead to changes in geometry, particularly the shortening of the bending arms, and, hence, errors in the computations. This leads to non-linear load deflection curves (curving upwards), and correction factors are needed. These can be calculated from large deflection beam theory (2) (3) and can be approximated, as described in Appendix 1, in the following forms

(i) DCB test

$$\frac{G}{G_0} = F = 1 - \frac{3}{10} \left(\frac{\delta}{a}\right)^2 - \frac{3}{2} \left(\frac{\delta}{a} \cdot \frac{l_1}{a}\right) \quad (8)$$

where δ is the load point deflection and G_0 is that value given in equation (4) for $h_1 = h_2 = h$, i.e.

$$G_0 = \frac{12}{B^2 E_{11}} \cdot \frac{P^2 a^2}{h^3}$$

For the assymetric case the factor of 3/10 may be replaced by

$$\frac{3}{5} \frac{\{1 + (h_1/h_2)^6\}}{\{1 + (h_1/h_2)^3\}^2}$$

It should be noted that large displacements in the assymetric case lead to mixed mode loading conditions since the thinner arm will shorten more than the thicker but the mode II component will be small for the usual range of (δ/a) , i.e., < 0.5 ,

(ii) ELS test

For the symmetrical test

$$G_0 = \frac{9}{4B^2} \frac{P^2 a^2}{E_{11} h^3}$$

and

$$\frac{G}{G_0} = F = 1 - \frac{\{15 + 50(a/L)^2 + 63(a/L)^4\}}{\{1 + 3(a/L)^3\}^2} \times \frac{3}{20} \left(\frac{\delta}{L}\right)^2 + \frac{3}{h} \cdot \frac{\delta}{L} \cdot \frac{l_1}{a} \frac{\{1 + 3(a/L)^2\}}{\{1 + 3(a/L)^3\}} \quad (9)$$

These corrections are generally larger than in the DCB case since for an equivalent case, say $\delta/a = 0.2$ in the DCB the factor is about 1 per cent while for the ELS test, $\delta/L = 0.1$, and for $a/L = 0.2$ the correction is 2 per cent, although there is some compensation from the second term.

(iii) AELS test

Here we again have a correction to G_0 which is given as either of equations (6) and

$$\frac{G}{G_0} = F = 1 - \frac{\{15 + (20\beta - 30)(a/L)^2 + (8\beta^2 - 20\beta + 15)(a/L)^4\}}{\{1 + (\beta - 1)(a/L)^3\}^2} \times \frac{3}{20} \left(\frac{\delta}{L}\right)^2 - 3 \cdot \left(\frac{\delta}{L}\right) \cdot \left(\frac{l_1}{a}\right) \frac{\{1 + (\beta - 1)(a/L)^2\}}{\{1 + (\beta - 1)(a/L)^3\}} \quad (10)$$

where

$$\beta = \left(\frac{h_1 + h_2}{h_2}\right)^3$$

It should be noted that in the limit of a very thin loaded arm with $\beta \rightarrow \infty$ we have

$$\frac{G}{G_0} = 1 - \frac{12}{5} \left(\frac{\delta}{a}\right)^2 - \frac{3}{2} \left(\frac{\delta}{a}\right) \left(\frac{l_1}{a}\right)$$

and clearly caution must be exercised in employing δ values which can be corrected since δ/a , and not δ/L , is the limiting factor.

A further factor to be considered is the fact that the shear modulus in laminates is much lower relative to the axial modulus than in isotropic materials. In the latter case the ratio of $E/G \approx 2.6$, while for laminates it may be up to 50. This leads to a much larger relative contribution of shear

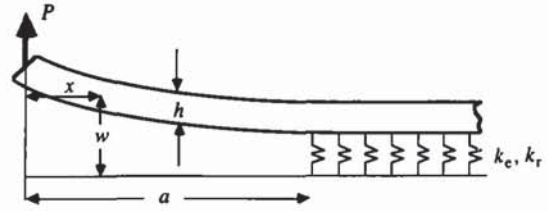


Fig. 4. Elastic foundation model

to the displacements and the necessary revision of correction factors. This is particularly marked in the built-in assumption used in the cantilever beams employed in these tests. An analysis has been given using a beam on elastic foundation model for the isotropic case (4) which can be extended to orthotropic materials (5). Figure 4 shows the model in which the deformation for $x > a$ is described by an elastic foundation of extensional stiffness k_e given by

$$k_e = \frac{2B}{h} \cdot E_{22}$$

and a rotational stiffness k_r given by

$$k_r = \frac{BhK}{2} \cdot G_{12}$$

where E_{22} is the transverse modulus and G_{12} the shear modulus and K is a stress distribution constant ≈ 0.85 .

The analysis includes the shear deformation by distinguishing between the slope of the neutral axis, dw/dx , and the angle of a normal plane ψ so that the shear strain is $(dw/dx - \psi)$. For $0 < x < a$ the beam equations become

$$E_{11} \frac{Bh^3}{12} \cdot \frac{d^3\psi}{dx^3} = E_{11} \frac{Bh^3}{12} \frac{d^4w}{dx^4} = 0$$

and the inclusion of shear deformations arises from the boundary conditions employed at $x = a$ which, for the conventional result, is achieved by placing $\psi = 0$ at $x = 0$. Since

$$\frac{-P}{Bh} = KG_{12} \left(\frac{dw}{dx} - \psi \right)$$

in this case dw/dx is determined and, hence, the shear corrected result

$$\delta = \frac{4Pa^3}{E_{11}Bh^3} + \frac{Pa}{KG_{12}Bh}$$

is obtained.

For the deformation when $x > a$ we have the equations

$$\left(\frac{E_{11}h^2}{12KG_{12}} \right) \frac{d^2\psi}{dx^2} = \frac{3\psi}{2} - \frac{dw}{dx}$$

and

$$\left(\frac{KG_{12}h^2}{2E_{22}} \right) \left(\frac{d^2w}{dx^2} - \frac{d\psi}{dx} \right) = w$$

These may be solved and by matching the displacement and slope at $x = a$ we include the shear in both parts of

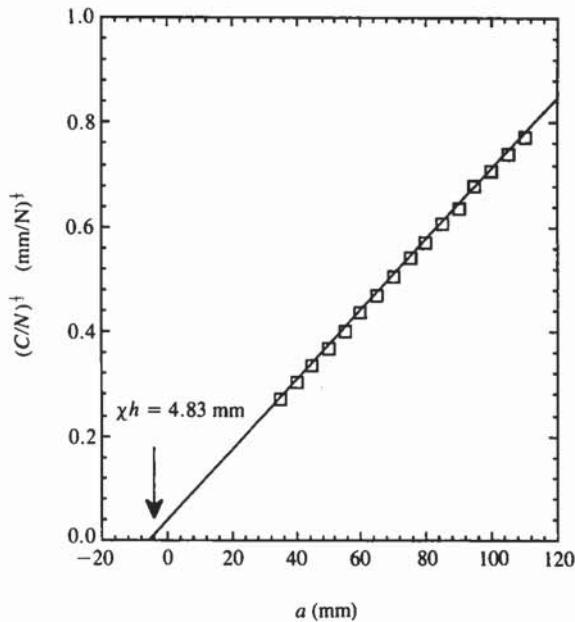


Fig. 5. Compliance calibration curve for AS4/PEEK composite

the beam. When this is done the conventional compliance of the cantilever from simple bending theory; i.e.

$$C = \frac{\delta}{P} = \frac{4a^3}{Bh^3E_{11}}$$

may be corrected for both shear deformation and root rotation by adding a length χh to a where

$$\chi = \left[\left(\frac{E_{11}}{11G_{12}} \right) \left\{ 3 - 2 \left(\frac{\Gamma}{1 + \Gamma} \right)^2 \right\} \right]^{1/2} \quad (12)$$

Table 1. Mode I data for AS4/PES

a (mm)	$E_{11, \text{uncorr}}$ (GPa)	N	$E_{11, \text{corr}}$ (GPa)	F	$P^2 a^2 / BEI$ (kJm ⁻²)	$3P\delta / 2Ba$ (kJm ⁻²)
30	80	0.932	128	0.955	1.62	1.62
35	83	0.944	126	0.953	1.67	1.69
40	87	0.949	125	0.952	1.81	1.83
45	91	0.951	125	0.950	1.91	1.94
50	97	0.954	130	0.949	1.95	1.91
55	100	0.954	129	0.948	2.05	2.01
60	100	0.954	127	0.947	1.98	1.98
65	106	0.954	131	0.946	2.05	1.97
70	111	0.954	130	0.945	2.05	2.00
75	111	0.952	131	0.943	2.05	1.98
80	111	0.951	130	0.941	1.98	1.94
85	112	0.948	130	0.937	2.03	1.98
90	110	0.943	126	0.931	2.06	2.09
95	113	0.941	128	0.928	2.13	2.12
100	112	0.934	124	0.921	2.22	2.27
105	114	0.929	124	0.915	2.31	2.36
110	114	0.924	124	0.909	2.45	2.43
115	118	0.921	126	0.905	2.47	2.48
120	117	0.916	124	0.900	2.41	2.47

(a) specimen dimensions: $2h = 4.05$ mm, $B = 20$ mm

(b) end block dimensions: $l_1 = 10$ mm, $l_2 = 7.01$ mm

(c) $E_{11} = 127$ GPa, $\chi = 2.92$

(d) N = correction factor for compliance

(e) F = correction factor for the large displacement and the end block

and

$$\Gamma = 1.18 \left[\frac{E_{11}E_{22}}{G_{12}} \right]^{1/2}$$

For unidirectional laminates $E_{11} \gg E_{22}$ and G_{12} so that $\Gamma \gg 1$ and

$$\chi \approx 0.3 \left\{ \frac{E_{11}}{G_{12}} + 3.4 \sqrt{\left(\frac{E_{11}}{E_{22}} \right)} \right\}^{1/2} \quad (13)$$

For isotropic materials $\chi \approx 0.7$, while for carbon fibre/polymer laminates typical values are $E_{11} = 130$ GPa, $E_{22} = 8$ GPa, and $G_{12} = 3.5$ GPa, giving $\chi = 2.4$; a much larger effect.

These corrections need to be made to the expression for G , i.e. equations (4), (8), and (6) in which a should be replaced by $(a + \chi h)$ in the first two and $(a + \chi h_2)$ in the third. In fact, when $h_1 \neq h_2$ there is a mixed mode component induced since each arm changes by a different amount, but it is a small effect.

It is of interest to note that corrections are often made via the compliance method using the form $C\alpha a^n$ and the curve fitting the data to find n . The best fit value of n to the $(a + \chi h)$ form is given when

$$n = 3 \left/ \left(1 + \frac{\chi h}{a} \right) \right.$$

and typically since $\chi h \approx 6$ mm for $h = 2$ mm and $a \approx 50 - 100$ mm, n will be about 2.7–2.8, as is observed. It is considered that the χh correction is a better representation since it models the actual mechanism involved.

4 MODULUS AND χ DETERMINATIONS

An important cross-check on consistency and a useful additional parameter is the axial modulus of the composite E_{11} which can be obtained from the same set of fracture data since δ , P , and a are known. It is most convenient to use the mode I DCB test for which the compliance expression is

$$C = \frac{\delta}{P} = \frac{8(a + \chi h)^3}{Bh^3E_{11}} \cdot N \quad (14)$$

where N is the finite displacement correction factor for both end blocks and finite displacement which may be deduced from equations (30)–(32) in Appendix 1. The result is

$$N = 1 - \left(\frac{l_2}{a} \right)^3 - \frac{9}{18} \frac{\delta l_1}{a^2} \left\{ 1 - \left(\frac{l_2}{a} \right)^2 \right\} - \frac{9}{35} \left(\frac{\delta}{a} \right)^2 \quad (15)$$

(Similar expressions may be deduced for the ELS and AELS specimens). E_{11} and χ may be found by plotting $(C/N)^{1/3}$ versus a from which χ may be found from the intercept at $\delta = 0$ and E_{11} from the slope (6). G_{IC} may then be found from equation (4); i.e.

$$G_{IC} = \frac{12P^2(a + \chi h)^2}{B^2E_{11}h^2} F \quad (16)$$

or alternatively E_{11} may be eliminated using equation (14) to give

$$G_{IC} = \frac{3}{2} \frac{P\delta}{B(a + \chi h)} \left(\frac{F}{N} \right) \quad (17)$$

Table 2. Mode I data for T300/epoxy

a (mm)	$E_{11, \text{uncorr}}$ (GPa)	N	$E_{11, \text{corr}}$ (GPa)	F	$P^2 a^2 / BEI$ (kJm ⁻²)	$3P\delta / 2Ba$ (kJm ⁻²)
40	88	0.959	128	0.984	0.27	0.27
50	94	0.973	126	0.985	0.26	0.26
55	98	0.977	128	0.984	0.27	0.27
60	99	0.979	127	0.984	0.26	0.27
65	102	0.981	128	0.984	0.26	0.26
70	103	0.982	127	0.984	0.26	0.26
75	105	0.983	128	0.984	0.27	0.27
80	108	0.984	130	0.984	0.26	0.26
85	109	0.984	130	0.984	0.27	0.27
90	110	0.985	129	0.983	0.27	0.26
95	109	0.985	128	0.983	0.25	0.25
100	111	0.985	129	0.983	0.27	0.26
105	109	0.985	126	0.982	0.26	0.26
110	112	0.985	129	0.982	0.26	0.26
115	113	0.985	128	0.981	0.26	0.26
120	114	0.984	129	0.982	0.26	0.26
125	113	0.984	127	0.981	0.25	0.26
130	114	0.984	128	0.981	0.26	0.26
135	114	0.984	127	0.980	0.26	0.26

 (a) specimen dimensions : $2h = 4.18$ mm, $B = 25$ mm

 (b) end block dimensions : $l_1 = 12.5$ mm, $l_2 = 7.045$ mm

 (c) $E_{11} = 128$ GPa, $\chi = 2.69$

 (d) N = correction factor for compliance

 (e) F = correction factor for the large displacement and the end block

Similar corrections can be worked out for the ELS and AELS specimens using the results of Appendix 1 (7).

5 EXPERIMENTAL RESULTS

Details of the use of these methods for the various test configurations will be given elsewhere (7) and here we will confine ourselves to data taken from three materials tested in mode I. Figure 5 shows a typical line obtained from a uniaxial AS4 carbon/PEEK composite where the corrected compliance (C/N) taken to the third root is shown as a function of crack length. As required by equation (14) there is good linearity indicating a constant modulus and correction length χh . This is typical of data from this type of test. Tables 1, 2, and 3 give the detailed results for three tests on AS4/PES, T300/epoxy, and AS4/PEEK, respectively. (These are all carbon fibres, the

Table 3. Mode I data for AS4/PEEK

a (mm)	$E_{11, \text{uncorr}}$ (GPa)	N	$E_{11, \text{corr}}$ (GPa)	F	$P^2 a^2 / BEI$ (kJm ⁻²)	$3P\delta / 2Ba$ (kJm ⁻²)
35	86	0.947	120	0.958	1.77	1.84
40	93	0.954	125	0.959	1.86	1.86
45	98	0.959	127	0.958	1.89	1.86
50	101	0.960	127	0.957	1.96	1.92
55	102	0.960	126	0.995	2.01	2.00
60	103	0.960	125	0.954	2.02	2.01
65	105	0.960	125	0.953	1.98	1.97
70	105	0.960	123	0.952	1.95	1.97
75	107	0.958	124	0.950	1.94	1.96
80	110	0.957	125	0.948	2.00	2.00
85	111	0.955	125	0.944	2.05	2.04
90	114	0.954	127	0.944	2.07	2.04
95	111	0.951	123	0.941	1.99	2.02
100	114	0.949	125	0.938	2.09	2.09
105	115	0.948	125	0.937	2.01	2.02
110	117	0.944	126	0.933	2.12	2.11

 (a) specimen dimensions : $2h = 4.37$ mm, $B = 20$ mm

 (b) end block dimensions : $l_1 = 10$ mm, $l_2 = 7.09$ mm

 (c) $E_{11} = 125$ GPa, $\chi = 2.21$

 (d) N = correction factor for compliance

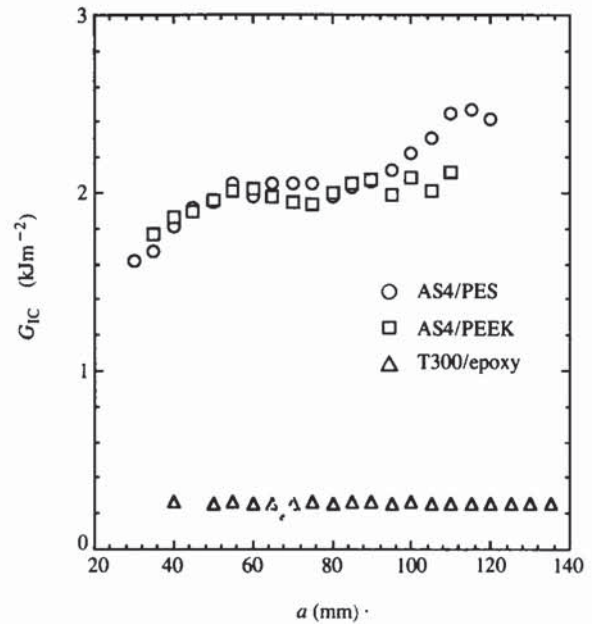
 (e) F = correction factor for the large displacement and the end block


Fig. 6. R curves for the three materials in Mode I

epoxy is a thermoset resin and PES (polyethersulphone) and PEEK (polyetheretherketone) are thermoplastics.) The correction factors N and F are all in the range 0.9–1. The second column gives the value of modulus without the χh correction and there are clearly substantial variations which are removed for the corrected value in the fourth column. The two values of G_{IC} given in the last two are derived from equation (16) and (17) and agree well demonstrating the accuracy for the corrected beam theory analysis. It should also be noted that the χ values agree well with those expected from the analysis, i.e., 2.2, 2.7, and 2.7, compared with 2.4 calculated.

Figure 6 shows the mode I data plotted as G_{IC} versus a and shows the constant, low value for epoxy, while the thermoplastic materials are much tougher, but show an increasing G_{IC} or R curve effect, although the AS4/PES does give a constant steady-state propagation value. This phenomenon again will be discussed in detail elsewhere (7). Finally, in Fig. 7 we have some data on one of the materials (AS4/PES) plotted as failure loci at crack initiation and at the steady state propagation value. The mode II data is from the ELS test and the mixed mode from the AELS version. The lines are drawn for the interaction model given in equation (3) and the form fits quite well with I increasing from 0.26 at initiation to 3.12 at propagation.

6 CONCLUSIONS

The examples given here show that the application of rather conventional elastic beam analysis, albeit augmented by large displacement effects, can provide accurate solutions for calculating the G_I and G_{II} components of the various test geometries. The mode I data given for these materials shows that the method is consistent and provides both toughness and modulus values. There are striking R curve effects in both thermoplastic matrix materials. The application of the analyses outlined give similarly reliable data in the other loading modes, and

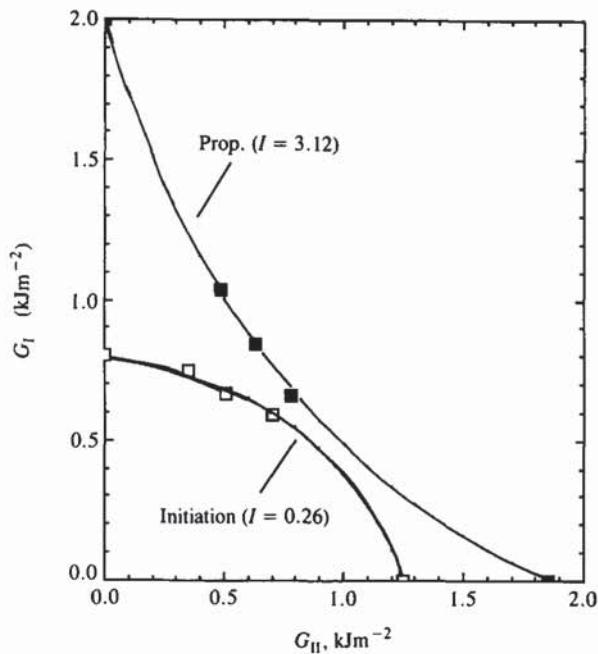


Fig. 7. The comparison between the failure loci at initiation and steady-state crack propagation in AS4/PES composite

the failure locus given shows that there is a rather complicated interaction of the various modes as the crack grows which can be characterised by the increasing interaction parameter, I . There is obviously much to be done to clarify such mechanisms, but it has been established that the methods of analysis are sound and form the basis for this further study.

APPENDIX 1

Corrections for the shortening of the loading arms may be computed from the large displacement analysis of beams (2) (3) which is conducted in terms of the angle of the beam slope ϕ . All cases of interest here may be deduced from the general cantilever shown in Fig. 8.

In the length a the second moment of area is I_2 and in the remainder, $L - a$, it is I_1 . Using the following parameters

$$\lambda^2 = \frac{2El_2}{P} \quad \text{and} \quad \beta = \frac{I_1}{I_2}$$

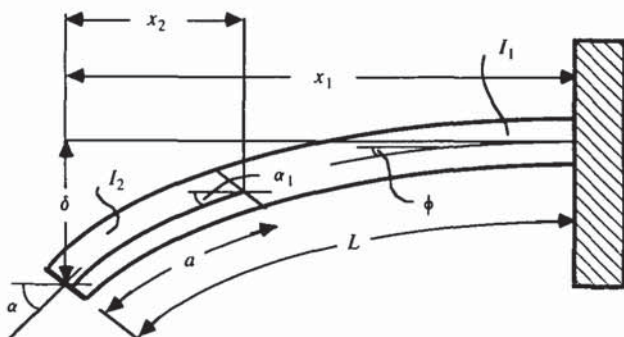


Fig. 8. Geometry and loading

we may formulate the solution of the problem as

$$\frac{a}{\lambda} = \frac{1}{2} \int_{\alpha_1}^{\alpha} \frac{d\phi}{\sqrt{(\sin \alpha - \sin \phi)}} \quad (18)$$

$$\frac{(L - a)}{\lambda} = \frac{1}{2} \int_0^{\alpha_1} \frac{\beta d\phi}{\sqrt{\{\sin \alpha + (\beta - 1) \sin \alpha_1 - \beta \sin \phi\}}} \quad (19)$$

$$\begin{aligned} \frac{\delta}{\lambda} = & \frac{1}{2} \int_{\alpha_1}^{\alpha} \frac{\sin \phi d\phi}{\sqrt{(\sin \alpha - \sin \phi)}} \\ & + \frac{1}{2} \int_0^{\alpha_1} \frac{\beta \sin \phi d\phi}{\sqrt{\{\sin \alpha + (\beta - 1) \sin \alpha_1 - \beta \sin \phi\}}} \end{aligned} \quad (20)$$

The shortened loading arm lengths are

$$\frac{x_1}{\lambda} = \sqrt{\{\sin \alpha + (\beta - 1) \sin \alpha_1\}} \quad (21)$$

$$\frac{x_2}{\lambda} = \sqrt{(\sin \alpha - \sin \alpha_1)} \quad (22)$$

These general results are intractable analytically but useful approximations may be obtained in terms of small angle approximations. Integrating equation (18) by parts we have

$$\begin{aligned} \frac{a}{\lambda} = & \sqrt{\left(\frac{\sin \alpha - \sin \alpha_1}{\cos \alpha_1} \right)} \\ & + \int_{\alpha_1}^{\alpha} \frac{\sin \phi}{\cos^2 \phi} \sqrt{(\sin \alpha - \sin \phi)} \cdot d\phi \end{aligned}$$

and substituting from equation (22) gives

$$\frac{x_2}{a} = \cos \alpha_1 \left\{ 1 - \frac{\lambda}{a} \int_{\alpha_1}^{\alpha} \frac{\sin \phi}{\cos^2 \phi} \sqrt{(\sin \alpha - \sin \phi)} \cdot d\phi \right\} \quad (23)$$

This may be approximated by considering the small angle solutions of equations (18), (19), and (20), i.e.

$$\frac{a}{\lambda} = \sqrt{(\alpha - \alpha_1)}, \quad \frac{L}{\lambda} = \sqrt{\{\alpha + (\beta - 1)\alpha_1\}},$$

$$\frac{\delta}{\lambda} = \frac{2}{3\beta} \frac{1}{\lambda^3} \{L^3 + (\beta - 1)a^3\} \quad (24)$$

and for equation (23)

$$\frac{x_2}{a} \approx 1 - \left(\frac{3\alpha_1^2 + 4\alpha_1\alpha + 8\alpha^2}{30} \right) \quad (25)$$

Similarly, for x_1

$$\begin{aligned} \frac{x_1}{L} \approx & 1 - \frac{2}{15\beta^2} \left[2\{\alpha + (\beta - 1)\alpha_1\}^2 + \left(\frac{a}{L} \right) (\beta - 1) \right. \\ & \left. \times (\alpha - \alpha_1) \{2(\alpha - \alpha_1) + \beta(2\alpha + 3\alpha_1)\} \right] \end{aligned} \quad (26)$$

From equation (24) we may derive expressions for α and α_1

$$\alpha = \left(\frac{3}{2} \frac{\delta}{L} \right) \left\{ \frac{1 + (\beta - 1)(a/L)^2}{1 + (\beta - 1)(\alpha/L)^3} \right\}$$

and

$$\alpha_1 = \left(\frac{3}{2} \frac{\delta}{L} \right) \left\{ \frac{1 - (a/L)^2}{1 + (\beta - 1)(a/L)^3} \right\} \quad (27)$$

and hence

$$\frac{x_2}{a} = 1 - \frac{3}{40} \left(\frac{\delta}{L} \right)^2 \frac{\{15 + (20\beta - 30)(a/L)^2 + (8\beta^2 - 20\beta + 15)(a/L)^4\}}{\{1 + (\beta - 1)(a/L)^3\}^2} \quad (28)$$

and

$$\frac{x_1}{L} = 1 - \frac{3}{10} \left(\frac{\delta}{L} \right)^2 \frac{\{2 + 5(\beta - 1)(a/L)^3 + (\beta - 1)(2\beta - 3)(a/L)^5\}}{\{1 + (\beta - 1)(a/L)^3\}^2} \quad (29)$$

Note that as $(a/L) \rightarrow 1$ both solutions give

$$\frac{x_2}{a} = \frac{x_1}{L} = 1 - \frac{3}{5} \left(\frac{\delta}{L} \right)^2$$

which is the same result as for x_1/L when $(a/L) \rightarrow 0$. At this limit

$$\frac{x_2}{a} \rightarrow 1 - \frac{9}{8} \left(\frac{\delta}{L} \right)^2$$

which is the small angle approximation of the exact solution for this limit

$$x_2 = a \cos \alpha$$

For the DCB test we need only the limit at $a/L = 1$ and if we take δ as the full deflection, then

$$\frac{x_2}{a} = 1 - \frac{3}{5} \left(\frac{\delta}{2a} \right)^2$$

and the correction in G , which is for a^2 becomes

$$1 - \frac{6}{5} \left(\frac{\delta}{2a} \right)^2 = 1 - \frac{3}{10} \left(\frac{\delta}{a} \right)^2$$

as given in equation (8). For the ELS mode test

$$\beta = \frac{(h_1 + h_2)^3}{h_1^3 + h_2^3}$$

in general and for the centrally cracked section $\beta = 4$ and we obtain equation (9) for correcting a^2 .

For the AELS test we have

$$\beta = \left(\frac{h_1 + h_2}{h_2} \right)^3$$

and hence equation (10). It should be noted that corrections for other configurations may be found from equation (25) by determining the appropriate expressions for α and α_1 .

It is also possible to determine the axial modulus of the beam via the deflection δ and so similar corrections are required. These may be deduced by integrating equation (20) by parts to determine the first term in a series expansion and substituting for α and α_1 from equations

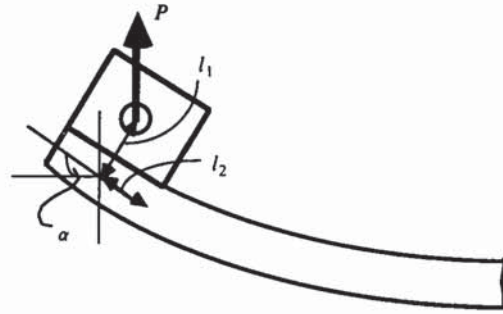


Fig. 9. End block

(27). The result may be written in the form

$$\delta = \frac{P}{3EI_1} \{L^3 + (\beta - 1)a^3\}N$$

where

$$N = 1 - \Theta \left(\frac{\delta}{L} \right)^2$$

and

$$\Theta = \frac{9}{70} \frac{[8 + (\beta - 1)(a/L)^3\{35 + 14(2\beta - 3) \times (a/L)^2 + (8\beta^2 - 20\beta + 15)(a/L)^4\}]}{\{1 + (\beta - 1)(a/L)^3\}^3} \quad (30)$$

Similar corrections are required if rigid end blocks are used to load the specimens shown in Fig. 9. The rotation of the block by α leads to a change in the moment arm of $l_1\alpha$ so that, using equation (27), we have an additional correction factor for a of

$$\frac{3}{2} \frac{\delta}{L} \cdot \frac{l_1}{a} \left\{ \frac{1 + (\beta - 1)(a/L)^2}{1 + (\beta - 1)(a/L)^3} \right\} \quad (31)$$

Similarly in the correction for δ , N , we have

$$\frac{\beta(l_2/L)^3}{\{1 + (\beta - 1)(a/L)^3\}} \quad (32)$$

which corrects for stiffening and

$$\frac{9}{4} \frac{\delta l_1}{L^2} \frac{[\{1 - (a/L)\}\{1 + (\beta - 1)(a/L)^3\} + \{1 - (l_2/a)^2\}\beta(a/L)^2\{1 + (\beta - 1)(a/L)^2\}]}{\{1 + (\beta - 1)(a/L)^3\}^2} \quad (33)$$

for the decrease in a and L .

APPENDIX 2

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